

THE BEHAVIOUR OF NORMAL PLOTS UNDER NON-NORMALITY FOR MIXED MODELS

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Summary: A general assumption in the analysis of linear mixed models is that the random effects and error terms have a Gaussian distribution. In this paper normal probability plots and formal hypothesis tests of normality are suggested and evaluated via a simulation study. The shapes of some of the probability plots suggest the transformation of the data which leads to responses that are close to normally distributed.

1. Introduction

We consider the linear mixed model

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \varepsilon$$

in which β is a $p \times 1$ vector of unknown constants (the fixed effects), $\mathbf{X}$ is an $n \times p$ design matrix of known numbers associated with β , $\mathbf{Z} = [\mathbf{Z}_1 | \mathbf{Z}_2 | \dots | \mathbf{Z}_r]$,

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where $\mathbf{Z}_i$ is an $n \times q_i$ design matrix for the random effect factor i , $\mathbf{u}' = [\mathbf{u}'_1 | \mathbf{u}'_2 | \dots | \mathbf{u}'_r]$, where $\mathbf{u}_i$ is a $q_i \times 1$ vector of independent random variables from $N(0, \sigma_i^2)$, $i = 1, 2, \dots, r$, ε is an $n \times 1$ vector of error terms from $N(0, \sigma_e^2)$ and $\mathbf{u}_i$ and ε are mutually independent. One may also write $\mathbf{u} \sim N(0, \mathbf{G})$, where $\mathbf{G}$ is block diagonal with i th block $\sigma_i^2 \mathbf{I}_{q_i}$, so that $\mathbf{y}$ has a multivariate normal distribution with $E(\mathbf{y}) = \mathbf{X}\beta$ and $\text{var}(\mathbf{y}) = \mathbf{V}$, where $\mathbf{V} = \sigma_e^2 \mathbf{I}_n + \mathbf{Z}\mathbf{G}\mathbf{Z}' = \sigma_e^2 \mathbf{I}_n + \sum_{i=1}^r \sigma_i^2 \mathbf{Z}_i \mathbf{Z}_i'$.

Given a sample vector, $\mathbf{y}$, the usual aim is to accomplish one or more of the following:

1. Estimate some linear functions of β .
2. Test hypotheses regarding β or functions of β .
3. Draw inferences on certain functions of the variance components.
4. Estimate $\mathbf{u}$ or some linear functions of $\mathbf{u}$.
5. Estimate jointly linear functions of β and $\mathbf{u}$.

The distributional assumption for these statistical inferences is that the error term and the random effects are normally distributed. The popular estimation methods for the linear mixed methods, maximum likelihood and restricted maximum likelihood, are based on the normality assumption.

However, data analysts often apply the mixed model mechanically to data which have both fixed and random effects without examining the appropriateness of the normality assumption. Using simulations and the analysis of a real data set, Butler and Louis (1992) have shown that an incorrect specification of the distribution of univariate random effects has little effect on the fixed effects estimates or on estimates of the residual variance and variance of the random effects. Verbeke and Lesaffre (1997) also analyzed two data sets with a linear mixed-effects model with random effects drawn

from a mixture of g multivariate normal distributions and found that varying the mixture had practically no effect on fixed effects estimates, but also that the residual variance and the covariance matrix of the random effects were very stable. Though Butler and Louis (1992) and Verbeke and Lesaffre (1997) studied the consequences of misspecifying the random effects assumption of normality, they did not address the issue of how to make a formal and/or informal assessment of the normality assumptions in linear mixed models. The objective of this paper is to explore formal and informal assessments of normality assumptions in linear mixed models.

2. Background

Since working in terms of $\gamma_i = \sigma_i^2/\sigma_e^2$ eases the computational procedures associated with our model, we write $\text{var}(\mathbf{y}) = \sigma_e^2 \mathbf{H}$, with $\mathbf{H} = \mathbf{I}_n + \mathbf{ZDZ}' = \mathbf{I}_n + \sum_{i=1}^r \gamma_i \mathbf{Z}_i \mathbf{Z}_i'$ and $\mathbf{D} = \mathbf{G}/\sigma_e^2$ (see Wolfinger et al., 1994 and Littell et al., 1996). The best linear unbiased estimate (BLUE) of β is $\hat{\beta} = (\mathbf{X}'\mathbf{H}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{H}^{-1}\mathbf{y}$ and the best linear unbiased predictor (BLUP) of $\mathbf{u}$ is $\tilde{\mathbf{u}} = \mathbf{DZRy}$ where $\mathbf{R} = \mathbf{H}^{-1} - \mathbf{H}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{H}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{H}^{-1}$. The maximum likelihood (ML) and restricted maximum likelihood (REML) estimates of σ_e^2 are $\hat{\sigma}_e^2 = \mathbf{y}'\mathbf{Ry}/n$ and $\hat{\sigma}_e^2 = \mathbf{y}'\mathbf{Ry}/(n-p)$, respectively. If the variance components γ_i are unknown, which is usually the case, the ML, REML or minimum norm quadratic unbiased estimation (MINQUE) estimates are simply substituted into the expressions. In this case the BLUE and BLUP acronyms no longer apply, but a qualifying "empirical" is often added to indicate such approximation. It is important to note that if the variance components or their estimates are available the (empirical) BLUE and (empirical) BLUP can be obtained with or without the assumption of normality (see, for instance, Robinson (1991) and Searle, Casella and

McCulloch (1992)). However, the normality assumption is necessary in order to make valid inferences.

After fitting the linear mixed model, the predicted values from the original data are $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} + \mathbf{Z}\tilde{\mathbf{u}}$ and the residuals (deviation of the responses from the fitted model) are $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{Z}\tilde{\mathbf{u}}$. This definition of residuals is used in most of the literature on linear mixed models; see, for example, Fellner (1986), Robinson (1991), SAS Institute (1992), Searle and Newsome-Stewart (1995) and Richardson (1997). Nevertheless, in the longitudinal mixed model there are two meaningful residual vectors, namely $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$, which helps to identify the subject outlier, and $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{Z}\tilde{\mathbf{u}}$ which helps to identify the case outlier within the subject or in the whole data set; see, for example, Jones (1993) and Verbeke (1997).

When Robinson (1991) criticized the use of $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ as residuals in the mixed model, he stressed that the model is then no longer viewed as a mixed model, but rather as a generalized linear regression. With the assumption that the error vector is normal with mean zero and variance $\sigma_e^2 \mathbf{H}$, the generalized least squares estimates are identical to the mixed model estimate of the parameters $\boldsymbol{\beta}$ and of the residuals. Thus, unless we use the random effects for prediction, why do we use the mixed model if the usual regression model gives us the same result?

Moreover, since for a linear combination $\mathbf{b}'\boldsymbol{\beta} + \mathbf{a}'\mathbf{u}$ such that $\mathbf{b}'\boldsymbol{\beta}$ is estimable, the best linear unbiased estimator is $\mathbf{b}'\hat{\boldsymbol{\beta}} + \mathbf{a}'\tilde{\mathbf{u}}$, we have for any given $\mathbf{y}$ that

$$\text{BLUP}(\boldsymbol{\varepsilon}) = \text{BLUP}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{Z}\tilde{\mathbf{u}} = \mathbf{e} = \mathbf{R}\mathbf{y};$$

see, e.g. Harville (1988, 1990), Robinson (1991), McLean Sanders and Stroup (1991) and Searle Casella and McCulloch (1992, Chapter 7)). Thus, the residual vector $\mathbf{e}$ is the best linear unbiased predictor of $\boldsymbol{\varepsilon}$. To sum up, since

the errors ε are unobservable, the assumptions about them must be checked indirectly using their BLUP $\mathbf{e}$.

From the model equation and the fact that $\mathbf{e} = \mathbf{R}\mathbf{y}$ and $\mathbf{R}\mathbf{X} = \mathbf{0}$, it follows that $\mathbf{e} = \mathbf{R}\mathbf{Z}\mathbf{u} + \mathbf{R}\boldsymbol{\varepsilon}$. Thus, the residuals may be used to assess the normality assumption, since if $\mathbf{e}$ is found to be non-normal, the normality of $\mathbf{u}$ or $\boldsymbol{\varepsilon}$, or both is suspect. Residuals are not independent and also are not directly comparable because they have different variances. The latter problem is largely obviated upon dividing each residual by an estimate of its standard error. Accordingly, the i th internally Studentized residual is $t_i = \frac{e_i}{\hat{\sigma}_e \sqrt{r_{ii}}}$.

To test the normality of the random effects vector $\mathbf{u}$ we could examine its BLUP. However, $\tilde{\mathbf{u}} = \mathbf{D}\mathbf{Z}'\mathbf{e}$, from which it follows that the predictor of the j th level of the i th random effect factor is γ_i times the sum of the residuals corresponding to the case where the j th level is matched. Therefore, the assessment of $\tilde{\mathbf{u}}$ is an assessment of a linear combinations of residuals. Consequently, it is sufficient to assess only the residuals.

3. Probability Plots and the W-statistic

The normal probability plot is obtained by plotting the order statistics of $t_1, \dots, t_n$ against $\alpha_i = \Phi^{-1}(\pi_i)$, for some choices of π_i , where Φ^{-1} denotes the inverse of the standard normal distribution function. Different values of π_i are used in practice. Mage (1982) reviewed nine plotting conventions recommended in the statistical literature. Cook and Weisberg (1982) noted that if $\pi_i = (i - 3/8) / (n + 1/4)$, then α_i provides an excellent approximation to the expected i th order statistic of a sample from the standard normal distribution. When the distribution is normal, the scatter plot appears roughly as a straight line through the origin and with unit slope. A non-negligible deviation from the straight line indicates violation of the normality

assumption. Probability plots can also be used to detect elements of the sample that differ from the others. Outlying elements of a sample will tend to fall toward the extremes of the plot, while most of the points will fall on the line that does not point toward the apparent outliers.

The W test statistic suggested by Shapiro and Wilk (1965) tests whether the sampled population is normal. The W statistic has been shown to be a good omnibus test of normality (Pearson, D'Agostino and Bowman (1977)). Suppose $t' = (t_1, t_2, \dots, t_n)$ is a random sample on which the W test of normality is to be carried out, let $b' = (b_1, b_2, \dots, b_n)$ be the vector of order statistics of t with $b_n = \max(t_i)$ and let $a' = (a_1, a_2, \dots, a_n)$ be the vector of expected values of standard normal order statistics. The W test statistic is defined as

$$W = \frac{(a' \Omega^{-1} b)^2}{a' \Omega^{-2} a \sum_{i=1}^n (t_i - \bar{t})^2}$$

where Ω is the variance-covariance matrix of the order statistics from the standard normal distribution. For $n \geq 6$, Royston (1982a) found by Monte Carlo simulation that

$$m = (1 - W)^\lambda$$

(where λ is a function of sample size, n) has an approximately normal distribution. He provided polynomials to evaluate λ , as well as $\mu(m)$ and $\sigma(m)$, the mean and standard deviation of m , for $7 \leq n \leq 2000$. This enables one to use $z = (m - \mu(m)) / \sigma(m)$ as a standard normal deviate for test purposes. Large values of z indicate departures from normality.

Royston (1982b, c) published algorithms (AS177 and AS181) embodying these methods for calculating the expected values of normal order statistics and the W-test for normality. The test statistic $z = (m - \mu(m)) / \sigma(m)$ is then referred to the upper tail of the $N(0, 1)$ distribution.

4. Simulation Study

The numerical study is performed using a mixed model simulation strategy analogous to that of Corbeil and Searle (1976) and Verbeke and Lesaffre (1997). As Beckman, Nachtsheim and Cook (1987) noted, in such an evaluation study using a simpler model has an advantage over a complex model. Their reasoning is that if the model is simple and the data conform exactly to expectation, we should be able to anticipate the consequence of a perturbation of the normality assumption. Moreover, Beckman et al. (1987) pointed out that in the development of any methodology it is always helpful to consider situations that allow for a relatively straightforward investigation of its operating characteristics. If the result behaves reasonably and can be justified by means of past experiences and general knowledge of the situation, then we have gained valuable insight and reassurance that the methodology will produce sensible results in more complicated settings. In this spirit, we use a mixed model with one or two fixed effects and one random effect and no interaction for the numerical study. We will examine both the balanced and unbalanced cases.

For the model,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}$$

we obtain the values of $\mathbf{X}$ and $\mathbf{Z}$ from our chosen examples, set $\boldsymbol{\beta} = \mathbf{0}$, generate random values from $\mathbf{u}$ and $\boldsymbol{\varepsilon}$ (by considering various distributional assumptions), and thus obtain $\mathbf{y}$. We also set the variance components (and hence, γ) to 1 (in fact any fixed value can be taken but this is chosen for simplicity in the data generation; see Butler and Louis, 1992 and Verbeke and Lesaffre, 1997). Khattree (1998) also noted that in the absence of any knowledge of the variance components, it is reasonable and customary to set to 1.

4.1 Unbalanced data set

One cannot cover all possible degrees of unbalancedness in any quantitative way. Instead one must choose intuitively designs that seem to represent fairly unbalanced designs; see Swallow and Searle (1978). Thus we selected the unbalanced design reported by Duchateau and Janssen (1997) on the effect of a drug on blood pressure. This example has a moderate sample size and a fairly unbalanced design.

In a small study, a new drug had been proven to be effective in decreasing blood pressure. In the next phase the drug was tested on a large scale. Nine different countries took part in the trial, and are considered to be a random selection from the set of all possible countries. The response variable is the measurement of difference in blood pressure (dbp) before and after treatment. There were different group sizes for different countries, shown in Table 1. In each group, each person was randomly assigned to drug or placebo. The main objective of the study was to see whether the conclusions of the small study can be confirmed.

The model is

$$y_{ijk} = \mu + \tau_i + c_j + \varepsilon_{ijk};$$

where, for $j = 1, \dots, 9$, $i = 1, 2$ and $k = 1, 2, \dots, n_{ij}$, y_{ijk} is the dbp of person k assigned to treatment i in country j , μ is the overall mean, τ_i is the main effect parameter for treatment i , c_j is the random effect of country j and ε_{ijk} is the random error corresponding to person k assigned to treatment i in country j .

Table 1. Numbers of patients in the sample, by treatment and country

Country	Drug	Placebo
Australia	10	6
Belgium	8	8
Canada	9	8
France	4	3
Germany	14	10
Great Britain	18	13
Korea	11	13
South Africa	22	21
Sweden	2	1
Total	98	83

4.2 Balanced data set

We use the cell proliferation design reported by Duchateau and Janssen (1997). Two drugs were tested in vitro for their ability to enhance cell proliferation in the presence and absence of a growth stimulating factor (gsf+ and gsf-). Nine plates were randomly selected from a population of plates. Each plate had two compartments and each compartment had two wells. First gsf+ and gsf- were randomly assigned to the two compartments of a given plate. Within a compartment, drugs A and B were randomly assigned to the two wells. All wells contained a cell culture with a fixed number of cells. The response is the number of cells in the well after a specified period of time.

The model equation is

$$y_{ijk} = \mu + \alpha_i + \beta_j + \eta_k + \varepsilon_{ijk};$$

where for $k = 1, \dots, 9$, $j = 1, 2$ and $i = 1, 2$, y_{ijk} is the logarithm of the cell count for gsf i , drug j and plate k , μ is the overall mean, α_i is the main effect parameter for gsf i , β_j is the main effect parameter for drug j , η_k is the random effect of plate k ; and ε_{ijk} is the random error for gsf i , drug j and plate k .

For both the balanced and unbalanced designs the random effect has 9 levels.

4.3 Distributional assumptions

We examine the effect of failure of the normality assumption by considering different distributions for the error term and the random effect. Namely, normal, $N(0, 1)$; lognormal, based on $N(0, 1)$; inverse normal, based on $N(0, 1)$; exponential with parameter 1; gamma with shape parameter 4; and uniform, $U(0, 1)$. In order to assess the behaviours of the probability plots for negatively skewed data, we considered pseudo-data which have a diametrically opposite skewness to the data generated from a gamma distribution with shape parameter 4. This is naively achieved by generating data from a gamma distribution with shape parameter 4 and then multiplying by -1 .

As Neter et al. (1996) noted, the analysis of model departures with respect to normality is particularly difficult when studying the nature of the probability distribution, unless the sample size is quite large. To investigate the use of the probability plot, when the actual distribution of our observed population is other than normal, we used the balanced design without replication and with 5 replications, as well as the unbalanced design with sample size 181.

4.4 Simulation results

All the simulations were conducted on a 600MHz PC using SAS macro programming, which iteratively calls the SAS MIXED procedure. Following Butler and Louis (1992) and Verbeke and Lesaffre (1997), ML estimation is used for all PROC MIXED calls. We repeatedly (1000 times) simulated data sets with different seeds from each of the above distributions and computed the W statistic and associated p -value. The simulated data sets were checked for possible spurious correlation, but none were found. The frequency

distributions of the p -values in the 1000 simulations are presented in Table 2. The categories of the p -values are less than or equal to 0.001, 0.001–0.05, 0.05–0.10, 0.10–0.25 and above 0.25. The entries in each cell are the percentage of times that the p -value for the W statistic lies in the above categories in the 1000 simulations.

Detection is taken as p -values falling below 0.05. For the balanced design with no replication, $n = 36$, which is fairly small, the departures from normality for the error term and for both the error term and the random effect term, are detected in over 83% of cases for the inverse normal, 80% of cases for the lognormal, but only 55% of cases for the exponential. For the gamma distributions these percentages fell into the low 20s (for both error term and random effects terms non-normal), while for the uniform distribution they fell to about 8%. However, departures from normality for the random effect term alone are seldom detected by the W statistic (less than 5.6% of the p -values are below 5%). As 5.6% is very close to the testing level of 5%, this essentially equals the false positive rate.

For the balanced design with 5 replications ($n = 180$) and the unbalanced design ($n = 181$), departures from normality are almost always detected, for all distributions, when the departure is in the error term or in both the random effect and error terms. It is never detected for departures from normality in the random effect term only.

The usefulness of graphical displays was examined using repeated simulations in the unbalanced and balanced designs. When looking at the plots, the nature of the graphs for the different simulations was similar. Thus we present only the probability plots for lognormal and gamma distributions in Figure 1. The entire set of plots are accessible through the web at <http://www.stats.unp.ac.za/staff/zewotir/mixedmodelprobabilityplots.pdf>

For sample size 36, visual inspection indicates that the plots are approximately linear with unit slope. However, a fairly wide band around the line of the slope is observed in some of the graphs. This is not particularly recognizable if the random effect is non-normal. But a spiral pattern is observed if the error term is non-normal.

For a small sample size, detection of non-normality using the probability plot is not simple. It requires experience – see, for instance, Cook and Weisberg, (1982). For the normal distribution, Daniel and Wood (1980) provided several pages of training plots that may help the analyst gain the necessary experience.

The uncertainty associated with small sample size probability plot inspection is alleviated to a certain extent at moderate or large sample sizes. When the hypothesized error term distribution is non-normal, the probability plot has an S shape and the W statistic is found to be significant. The change in the random effect distribution does not affect the linearity of the plot, and this is strongly supported by the test of normality based on W. It is also important to note that the nature and the pattern of each plot are in some ways similar to the corresponding plots for different sample sizes. But the plots are clearer and simpler to judge at increased sample sizes. In general, when only the distribution of the error term is non-normal, plots tend to follow specific non-linear patterns. But if only the distribution of the random effects is non-normal, the probability plot appears as a straight line. And these plots are almost identical to the probability plot obtained from normal random effect and normal error terms. Moreover, the probability plot obtained from a

Table 2. The percentage frequency distribution of the p-value of the W statistic under different distributions and designs

Distribution		Balanced design with no replication (n= 36)					Balanced design with 5 replications (n= 180)					Unbalanced design (n= 181)				
u	ϵ	Below 0.001	0.001 to 0.05	0.05 to 0.10	0.10 to 0.25	above 0.25	below 0.001	0.001 to 0.05	0.05 to 0.10	0.10 to 0.25	above 0.25	below 0.001	0.001 to 0.05	0.05 to 0.10	0.10 to 0.25	above 0.25
Normal	Normal	0.0	5.6	3.6	13.2	77.6	0.0	5.2	5.3	15.3	74.2	0.0	4.9	4.7	16.7	73.7
Normal	Lognormal	57.5	25.4	4.2	6.9	6.0	100	0.0	0.0	0.0	0.0	100	0.0	0.0	0.0	0.0
Normal	Inverse normal	84.8	8.4	1.3	2.0	3.5	100	0.0	0.0	0.0	0.0	100	0.0	0.0	0.0	0.0
Normal	Exponential	18.0	36.7	10.3	14.2	20.8	100	0.0	0.0	0.0	0.0	100	0.0	0.0	0.0	0.0
Normal	Gamma	4.0	22.0	10.0	19.5	44.5	88.6	10.0	0.6	0.8	0.0	90.9	8.2	0.7	0.2	0.0
	(+vely skewed)															
Normal	Gamma	5.6	24.8	9.5	18.8	41.1	87.8	10.9	0.7	0.4	0.2	89.1	10.2	0.2	0.3	0.2
	(-vely skewed)															
Normal	Uniform	0.1	8.5	7.7	21.8	61.9	96.3	3.6	0.1	0.0	0.0	97.5	2.4	0.0	0.1	0.0
Lognormal	Normal	0.2	4.2	6.7	13.3	75.6	0.1	4.5	6.0	16.0	73.4	0.2	5.4	4.9	15.0	74.5
Inverse normal	Normal	0.0	3.5	4.4	14.5	77.6	0.3	5.1	5.9	14.9	73.8	0.1	5.1	5.1	14.6	75.1
Exponential	Normal	0.1	4.3	4.3	15.3	76.0	0.0	5.1	5.8	14.7	74.4	0.3	4.9	6.3	14.6	73.9
Gamma	Normal	0.1	4.8	4.8	14.8	75.5	0.1	5.0	4.5	15.4	75.0	0.0	5.0	3.6	14.3	77.1
(+vely skewed)																
Gamma	Normal	0.1	5.4	4.1	15.8	74.6	0.1	6.1	4.6	14.5	74.7	0.1	4.4	5.5	17.2	72.8
(-vely skewed)																
Uniform	Normal	0.1	3.0	4.3	14.9	77.7	0.2	5.2	6.0	13.5	75.1	0.0	5.3	5.0	13.2	76.5
Lognormal	Lognormal	50.8	30.0	4.3	6.7	8.2	100	0.0	0.0	0.0	0.0	100	0.0	0.0	0.0	0.0
Inverse normal	Inverse normal	68.9	14.4	1.8	5.4	9.5	100	0.0	0.0	0.0	0.0	100	0.0	0.0	0.0	0.0
Exponential	Exponential	21.0	38.3	9.3	12.4	19.0	100	0.0	0.0	0.0	0.0	100	0.0	0.0	0.0	0.0
Gamma	Gamma	3.9	18.5	9.2	18.9	49.5	87.3	11.5	0.5	0.4	0.3	91.9	7.1	0.5	0.5	0
(+vely skewed)	(+vely skewed)															
Gamma	Gamma	2.7	18.4	10.1	17.3	51.5	87.2	11.0	1.0	0.6	0.2	90.2	8.7	0.6	0.3	0.2
(-vely skewed)	(-vely skewed)															
Uniform	Uniform	0.2	10.1	8.4	22.1	59.2	95.9	4.0	0.1	0.0	0.0	97.4	2.6	0.0	0.0	0.0

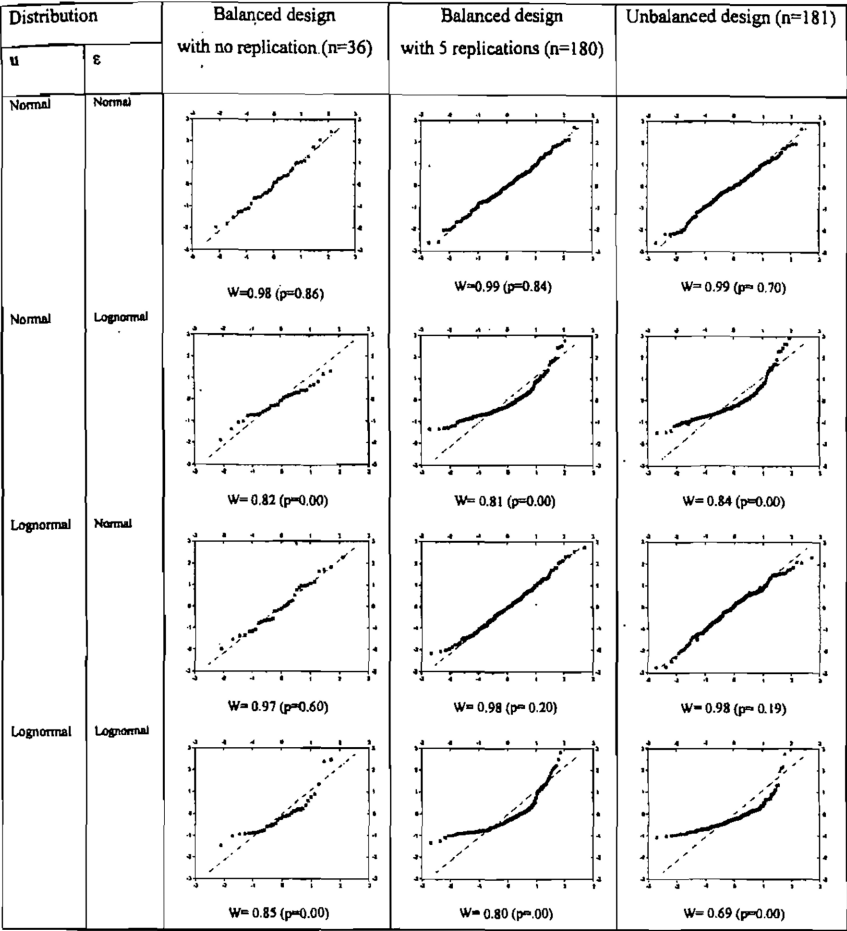


Figure 1. Probability plot of Studentized residual versus normal quantiles under lognormal and gamma-distributions in the random effect and error term

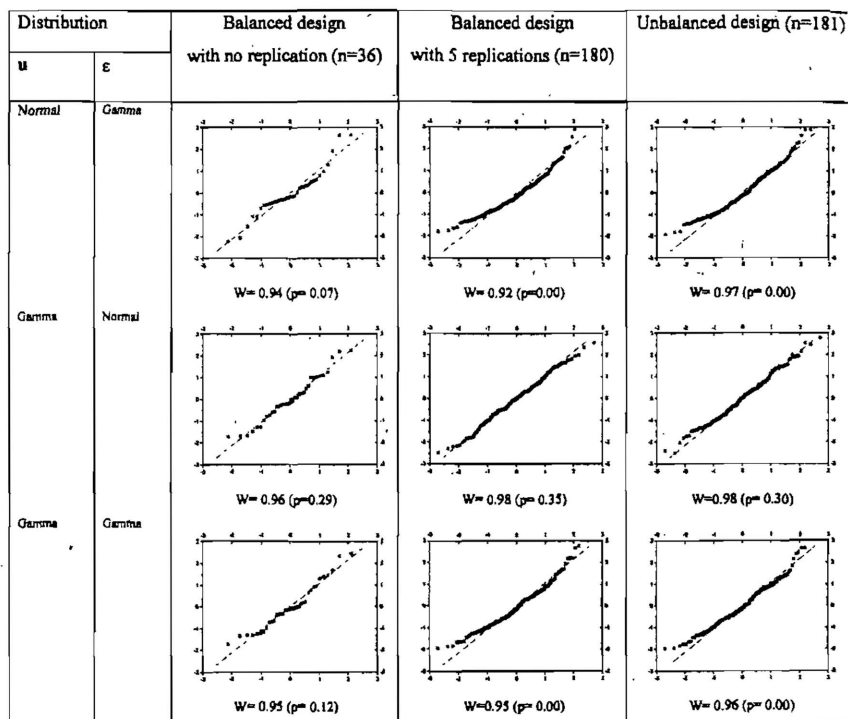


Figure 1. Probability plot of Studentized residual versus normal quantiles under lognormal and gamma distributions in the random effect and error term

non-normal random effect and error term is almost identical to the probability plot obtained from the corresponding non-normal error terms and normal random effect. This leads us to conclude that the assumption of normality in the random effect terms is irrelevant as long as the assumption of normality in the error terms is correct. In general, when only the distribution of the error concludes that the assumption of normality in the random effect terms is irrelevant as long as the term is non-normal, plots tend to follow specific nonlinear patterns. But if only the distribution of the random effects is non-normal, the probability plot appears as a straight line. And these plots are almost identical to the probability plot obtained from normal random effect and normal error terms. Moreover, the probability plot obtained from a non-normal random effect and error term is almost identical to the probability plot obtained from the corresponding non-normal error terms and normal random effect. This leads us to conclude that the assumption of normality in the random effect terms is irrelevant as long as the assumption of normality in the error terms is correct.

Obviously the shape of the probability plot depends on the distribution of the parent error term, hence on the distribution of the response variable. The plots indicate that relatively skewed distributions (exponential and Gamma) result in j-shaped or inverted j-shaped plots. For short-tailed distributions such as the uniform, the probability plot tends to be S-shaped. For the inverse normal distribution the probability plots have a tendency to be '~'-shaped. For the lognormal the plot tends to be a wide upward parabola. Since the inverse normal and lognormal can easily be transformed to normal, transforming the data would be the solution for the violation of normality assumption. That is, if the shape of the probability plot looks '~'-shaped the inverse (reciprocal)

transformation may solve the problem. If the shape looks like a very wide upward parabola, a log transformation may be the remedy.

5. Aerosol penetration example

Beckman et al. (1987) developed local influence diagnostics for mixed model analyses and illustrated their methodology by using the aerosol-penetration data. Christensen, Pearson and Johnson (1992) used the same data for comparison and illustration of their diagnostic methods.

The aerosol data involves high-efficiency particulate air filter cartridges that are used in commercial respirators to prevent or reduce the respiration of toxic fumes, radionuclides, dusts, mists, and other particulate matter. The goals of the analysis are to determine factors that contribute most to the variability in the penetration of the filters and to determine whether the standard aerosol can be replaced by an alternative aerosol in quality assurance testing. Filters fail a quality test if the percentage of penetration of an aerosol is too large. Two aerosols were crossed with two filter manufacturers. Within each manufacturer, three filters were used to evaluate the penetration of the two aerosols (that is, a total of 6 filters were used). The data are presented in Table 3.

The aerosols and manufacturers were taken as fixed effects and filters nested within manufacturers were taken as a random effect. This led to the model

$$y_{ijkm} = \mu + \alpha_i + \beta_j + \eta_{jk} + \varepsilon_{ijkm}$$

where y_{ijkm} is the percent penetration, α_i is a fixed effect for the i th aerosol type, β_j is a fixed effect for the j th filter manufacturer, η_{jk} is a random effect for the k th filter nested within the j th manufacturer, and ε_{ijkm} is the error associated with the m th replication in the ijk th cell. The assumptions are

that the η_{jk} are iid $N(0, \sigma_1^2)$, the ε_{ijkm} are iid $N(0, \sigma_0^2)$ and $\{\eta_{jk}\}$ and $\{\varepsilon_{ijkm}\}$ are independent.

Table 3. Percent penetration of two aerosols using three filters from manufacturers

	Manufacturer 1 ($j = 1$)			Manufacturer 2 ($j = 2$)		
	Filter 1 ($k=1$)	Filter 2 ($k=2$)	Filter 3 ($k=3$)	Filter 4 ($k=4$)	Filter 5 ($k=5$)	Filter 6 ($k=6$)
Aerosol 1 ($i=1$)	0.750	0.082	0.082	0.600	4.100	1.000
	0.770	0.085	0.076	0.680	5.000	1.800
	0.840	0.096	0.077	0.870	1.800	2.700
	1.120	0.160	0.150	0.910	0.660	2.170
	1.100	0.110	0.120	0.830	0.830	1.520
	1.120	0.260	0.120	0.950	0.610	1.580

The normal probability plot in Figure 2 indicates that the assumption of normality is doubtful. The normality goodness-of-fit summary statistic is $W=0.84$. The plot and the test of normality based on W indicate violation of the normality assumption ($p = 0.0001$). It is possible that this violation of normality could arise from the apparent outliers. To check the outlier effect, we reassessed the normality when cases 14 ($i = 1, j = 2, k = 2$ and $m = 2$) and 13 ($i = 1, j = 2, k = 2$ and $m = 1$) are deleted. The summary statistics when cases 14, 13 and both 14 and 13 are deleted are $W = 0.79$ ($p < 0.0001$), $W = 0.73$ ($p < 0.0001$) and $W = 0.75$ ($p < 0.0001$), respectively. Thus, there is no change in the conclusion from the test of normality when outliers are individually or collectively omitted. This suggests that the data are from a long-tailed distribution.

The pattern of the probability plot tends to be a very wide concave upward parabola, indicating that a log transformation is appropriate. We refit the model after making a logarithmic transformation of the response variable (the

percent penetration). Then the probability plot (see Figure 3) is linear and $W = 0.96$ for which $p = 0.27$.

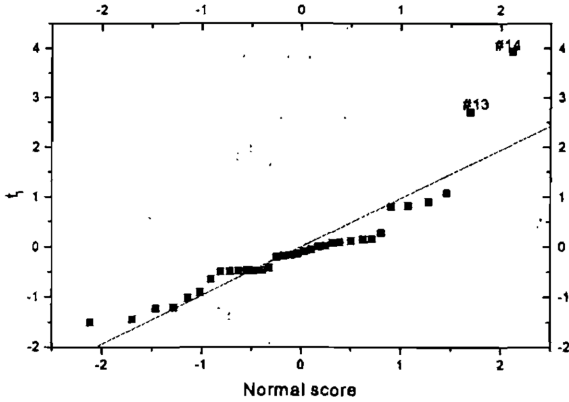


Figure 2. Aerosol data: Normal probability plot

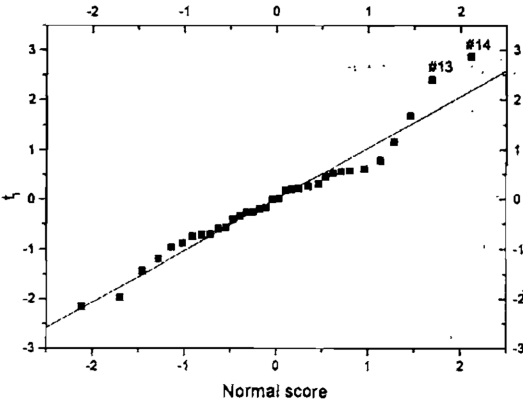


Figure 3. Aerosol data: Normal probability plot for log transformed fit

The summary results before and after transformation are presented in Table 4. The normal probability plot (Figure 3) does not indicate a serious violation of normality, although cases 14 and 13 appear to dominate our perception of this plot. The summary statistic $W = 0.96$ also favours normality ($p = 0.27$). The model fit information obtained from -2 times the log likelihood, Akaike's Information Criterion (AIC), AICC-a finite sample correction of AIC, and the Bayesian Information Criterion (BIC) show that the transformed model is very much better for these data. Cases 14 and 13 still appear as possible outliers. However, since the W test does not detect any significant deviation from normality, it appears that these are merely points slightly different to the others.

Although in this example there is no remarkable change in the significance of the fixed and random effects, there is a considerable change in the p -values. The response variable in the transformed model is in the log scale. In both models it can be concluded that there is no variability among all levels of filter (including those not sampled), there is no difference in response means between the two aerosol types, but there is a difference in response means between the two filter manufacturers at the 5% level of significance.

6. Conclusion

For a small sample size, interpretation of probability plots is not simple. Using the deviation from a straight line through the origin with slope one as an indication of the violation of normality, inspection of the band and the pattern is very helpful. A wide band around the line of the slope and distinct patterns that stand out on the plot are also good indicators of the violation of normality. Though such an examination is sufficient for many practical purposes, we suggest that a better approach is to augment such examination with a formal

hypothesis test based on the plot. Besides providing an objective criterion for interpreting the plot, such a test also provides a useful summary of information contained in the plot.

Table 4. Estimate of the parameters and model fit information from untransformed and transformed model

	Before transformation	After transformation
Filter effect	0.136 ($p = 0.34$)	0.531 ($p = 0.11$)
Fixed effects F -value		
Aerosol	2.21 ($p = 0.15$)	0.04 ($p = 0.85$)
Manufacturer	8.85 ($p = 0.01$)	7.96 ($p = 0.01$)
W	0.84 ($p = 0.00$)	0.96 ($p = 0.27$)
-2log likelihood	90.7	64.4
(smaller is better)		
AIC (smaller is better)	100.7	74.4
AICC (smaller is better)	102.7	76.4
BIC (smaller is better)	99.6	73.3

The simulation study showed that the probability plot is hardly affected by non-normality of the random effects. The plot seems sensitive to the distribution of the error terms only. This restricted sensitivity leads us to conclude that it is only the normality of the error term that is important in linear mixed model analysis. This conclusion is in agreement with Butler and Louis (1992) and Verbeke and Lesaffre (1997) who examined the failure of the normality assumption for random effects. They concluded that most of the optimum properties of the maximum likelihood estimators of linear mixed model parameters continue to hold when the random effects distribution is non-normal. These authors did not examine the failure of the assumption for the error term, or for both the error and random effects terms simultaneously.

The shape of the probability plot reflects the distribution of the error term in the model. Since a straight-line probability plot is an indication that

the underlying distribution is normal, we tried to associate some non-linear shapes of probability plots to known sampling distributions, thus suggesting an appropriate transformation to normality. Accordingly, we found that if the probability plot has a ‘~’-shape, an inverse normal transformation can be a remedy. If the shape looks like a very wide parabola, a logarithmic transformation may be useful. Otherwise inverted *j*-shaped plots indicate that the errors come from a skewed distribution. For short-tailed distributions, the probability plots tend to be S-shaped.

We do not claim that transformation is the only way of accommodating heavy-tailed distributions in a linear mixed model framework. Apparently heavy-tailed distributions are sometimes merely the result of genuine outliers in the data, in which case correction of the outliers may be the solution. The generalized linear mixed model approach of McCulloch and Searle (2001) is also an alternative remedy.

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